

Data Catalog: Sales Data Mart (Gold Layer)

Scope: **dim_product** and **fact_sales**

1. Overview

A star schema for sales analytics: one fact table (**fact_sales**) linked to the product dimension (**dim_product**).

Object	Type	Grain	Purpose
dim_product	Dimension	One row per product	Product attributes: category hierarchy, product line, cost
fact_sales	Fact	One row per product on an order (order line)	Sales transactions: dates, quantity, price, sales amount

Note: descriptions, data types and examples are inferred from column names. Adjust them to match your actual DDL and source systems.

2. Table: dim_product

Description: Descriptive attributes for every product sold, including its category hierarchy.

Primary key: product_key (surrogate)

Column	Suggested Type	Key	Description	Example
product_key	INT	PK (surrogate)	Surrogate key generated in the warehouse; used to join to fact_sales	1
product_id	INT	Business key	Product identifier from the source system	210
product_number	NVARCHAR(50)	Business key	Structured alphanumeric product code	BK-R93R-62
product_name	NVARCHAR(100)		Descriptive product name	Road-650 Red, 62
category_id	NVARCHAR(20)		Identifier of the product's category	CO_RF
category	NVARCHAR(50)		Top-level classification	Bikes
sub_category	NVARCHAR(50)		Second-level classification within the category	Road Bikes
product_line	NVARCHAR(50)		Product line or series the product belongs to	Road
cost	DECIMAL(18,2)		Base cost of the product	686.00
maintenance	NVARCHAR(10)		Flag indicating whether the product requires maintenance	Yes
start_date	DATE		Date the product became available / its record became effective	2012-07-01

3. Table: fact_sales

Description: Transactional sales data, one row per product line on an order.

Grain: One row per order_number + product_key

Foreign keys: product_key -> dim_product; customer_key -> dim_customer (not covered in this catalog)

Column	Suggested Type	Key	Description	Example
order_number	NVARCHAR(50)	Degenerate dimension	Unique order identifier from the source system	SO43697
product_key	INT	FK -> dim_product.product_key	Product sold on this order line	25
customer_key	INT	FK -> dim_customer.customer_key	Customer who placed the order	21768
order_date	DATE		Date the order was placed	2010-12-29
ship_date	DATE		Date the order was shipped	2011-01-05

Column	Suggested Type	Key	Description	Example
due_date	DATE		Date payment for the order was due	2011-01-10
sales	DECIMAL(18,2)	Measure	Total sales amount for the line (quantity x price)	3578.00
quantity	INT	Measure	Number of units sold	1
price	DECIMAL(18,2)	Measure	Price per unit	3578.00

4. Relationships

From (many)	To (one)	Join	Cardinality as drawn
fact_sales.product_key	dim_product.product_key	fact_sales.product_key = dim_product.product_key	One-or-many on both ends

dim_product ----< fact_sales

5. Business Rules and Notes

- **Surrogate vs. business keys:** Always join facts to dimensions on the _key columns, never on _id or _number.
- **Derived columns:** sales should equal quantity x price.
- **Date sequence:** order_date <= ship_date; due_date is expected after order_date.
- **Naming issue:** dim_product.maintenance is misspelled in the model (should be maintenance). Rename it before it spreads into downstream queries.
- **Relationship cardinality:** The diagram shows one-or-many at both ends of the dim_product to fact_sales link. In a star schema this is normally one-to-many (one product, many sales rows). Confirm that is what you intend.

6. Sample Analytical Query

```
-- Revenue by product category
SELECT p.category, SUM(f.sales) AS total_sales
FROM fact_sales f
JOIN dim_product p ON f.product_key = p.product_key
GROUP BY p.category
ORDER BY total_sales DESC;
```